

BRYAN KUMARA

<https://brykumara.github.io/>

PERSONAL INFORMATION

Place of Birth: Singapore

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BIOGRAPHY

I am a D.Phil. student at the University of Oxford where I am advised by Prof. Kasper Rasmussen researching applied cryptography for distributed systems. Before Oxford, I was a cryptographer at CrossBar Inc. working on threshold ECDSA and a researcher at the Alan Turing Institute. In the latter, I co-authored the Mahi-Mahi asynchronous DAG consensus protocol, a batched vector commitment for MPCitH-type signatures, and guidelines for implementing homomorphic encryption.

EXPERIENCES

- 2025 - Applied Cryptographer, CrossBar Inc., United States
Threshold ECDSA and Post-Quantum Cryptography
- 2023 - 2025 Research Assistant, The Alan Turing Institute, United Kingdom
Asynchronous Consensus, Batched Vector Commitments, Homomorphic Encryption
- 2021 - 2021 Project Management Intern, PayPal, United Kingdom
EU Exit Transition for PayPal UK
- 2019 - 2019 Undergraduate Researcher, Imperial College London, United Kingdom
Model Checking for Probabilistic Multi-Agent Systems
- 2018 - 2018 Project Intern, Boston Consulting Group, Indonesia
Timber and Tissue Paper Manufacturing in Indonesia
- 2017 - 2017 Research Assistant, The Habibie Center, Indonesia
Cybersecurity Policies in Asean

EDUCATION

- 2025 - University of Oxford, Oxford, United Kingdom
D.Phil in Computer Science
- 2021 -2022 University College London, London, United Kingdom
M.Sci in Emerging Digital Technologies
- 2017 - 2020 London School of Economics, London, United Kingdom
B.Sci in Philosophy, Logic, and Scientific Method

PUBLICATIONS, REPORTS, AND AWARDS

My full list of publications, technical reports, and awards is available at <https://brykumara.github.io/>